

Elektrotehnički fakultet Univerziteta u Beogradu

Principi Softverskog Inženjerstva (SI3PSI)

tikimiki

Specifikacija Baze Podatka - Verzija 1.1

digitalci

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1. Uvod

1.1. Namena dokumenta

Baza podataka za projekat tikimiki predstavlja fleksibilan i pouzdan način čuvanja podataka i pristupa istim od strane veb servera radi generisanja veb strana i pružanja funkcionalnosti platforme. U dokumentu se nalazi dijagram sa IE notacijom, šema relacione baze podataka, kao i opis svih tabela u bazi podataka. Ovaj dokument služi kao osnova za razvoj detaljne projektne specifikacije sistema, implementaciju i testiranje.

1.2. Ciljna grupa

Dokument je namenjen vodi (tim lideru) projekta i članovima razvojnog tima. Tim lideru ovaj dokument služi za planiranje razvojnih aktivnosti i specifikaciju imena tabela i polja u bazi, kako bi nezavisne celine, implementirane od strane različitih delova razvojnog tima, na kraju rada bile uspešno integrisane. Razvojnog timu dokument služi kao osnova za dizajn i implementaciju baze podataka.

1.3. Rečnik pojmova i skraćenica

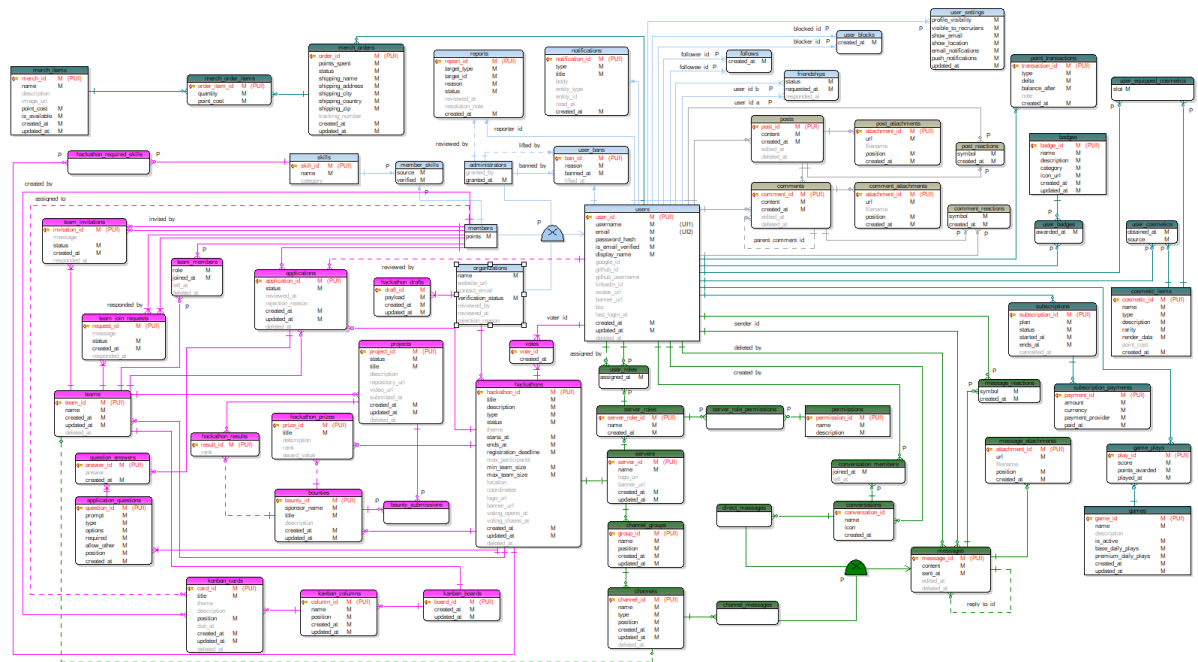
- IE – *Information Engineering*, notacija za modelovanje podataka

1.4. Otvorena pitanja

Broj	Opis	Rešenje

2. Modeli podataka

2.1. Dijagram IE notacije



2.2. Šema relacije baze podataka

```
users (user_id, username, display_name, email, password_hash,  
is_email_verified, google_id, github_id, github_username,  
github_access_token, linkedin_id, avatar_url, banner_url, bio,  
last_login_at, created_at, updated_at, deleted_at)
```

administrators (user_id, granted_by, granted_at)**members** (user_id, points)

organizations (user_id, name, website_url, logo_url, contact_email, verification_status, reviewed_by, reviewed_at, rejection_reason)

follows (follower_id, followee_id, created_at)

```
friendships (user_id_a, user_id_b, requester_id, status, requested_at,  
responded_at)
```

```
user_bans (ban_id, user_id, banned_by, reason, banned_at, lifted_at,
lifted_by)
```

```
follows (follower_id, followee_id, created_at)
```

```
friendships (user_id_a, user_id_b, requester_id, status, requested_at,  
responded_at)
```

```
user_bans (ban_id, user_id, banned_by, reason, banned_at, lifted_at,
lifted_by)
```

```
user_settings (user_id, profile_visibility, visible_to_recruiters,  
show_email, show_location, email_notifications, push_notifications,  
updated_at)
```

```
user_blocks (blocker_id, blocked_id, created_at)
```

skills (skill_id, name, category)
member_skills (user_id, skill_id, source, verified)
hackathons (hackathon_id, organization_id, title, description, type, status, theme, starts_at, ends_at, registration_deadline, voting_opens_at, voting_closes_at, max_participants, min_team_size, max_team_size, location, coordinates, logo_url, banner_url, created_at, updated_at, deleted_at)
hackathon_drafts (draft_id, organization_id, payload, created_at, updated_at)
hackathon_required_skills (hackathon_id, skill_id)
bounties (bounty_id, hackathon_id, sponsor_name, title, theme, description, created_at, updated_at)
hackathon_prizes (prize_id, hackathon_id, bounty_id, sponsor_name, title, description, rank, award_value)
teams (team_id, hackathon_id, name, created_at, updated_at, deleted_at)
team_members (team_id, user_id, role, joined_at, left_at, deleted_at)
team_join_requests (request_id, team_id, user_id, message, status, created_at, responded_at, responded_by)
team_invitations (invitation_id, team_id, user_id, invited_by, message, status, created_at, responded_at)
applications (application_id, user_id, hackathon_id, team_id, status, reviewed_by, reviewed_at, rejection_reason, created_at, updated_at, deleted_at)
application_questions (question_id, hackathon_id, prompt, type, options, required, allow_other, position, created_at)
question_answers (answer_id, application_id, question_id, answer, created_at)
projects (project_id, team_id, status, title, description, repository_url, video_url, submitted_at, created_at, updated_at, deleted_at)
hackathon_results (result_id, project_id, bounty_id, rank, prize_id)
bounty_submissions (bounty_id, project_id)
votes (vote_id, hackathon_id, project_id, voter_id, voter_fingerprint, created_at)
kanban_boards (board_id, team_id, created_at, updated_at)
kanban_columns (column_id, board_id, name, position, created_at, updated_at)
kanban_cards (card_id, column_id, created_by, assigned_to, title, description, position, due_at, created_at, updated_at, deleted_at)
servers (server_id, hackathon_id, name, logo_url, banner_url, created_at, updated_at)
server_roles (server_role_id, server_id, name, created_at)
permissions (permission_id, name, description)
server_role_permissions (server_role_id, permission_id)
user_roles (server_role_id, user_id, assigned_by, assigned_at)

channel_groups (group_id, server_id, name, position, created_at, updated_at)
channels (channel_id, group_id, team_id, type, name, position, created_at, updated_at, deleted_at)
channel_members (channel_id, user_id, added_by, added_at)
channel_pins (channel_id, message_id, pinned_by, pinned_at)
messages (message_id, sender_id, reply_to_id, content, sent_at, edited_at, deleted_at, deleted_by)
message_attachments (attachment_id, message_id, url, filename, position, created_at)
channel_messages (message_id, channel_id)
conversations (conversation_id, created_by, name, icon, created_at)
conversation_members (conversation_id, user_id, joined_at, left_at, last_read_at)
direct_messages (message_id, conversation_id)
message_reactions (user_id, message_id, symbol, created_at)
server_mutes (mute_id, server_id, muted_user_id, muted_by, muted_at, expires_at, reason)
posts (post_id, user_id, content, created_at, edited_at, deleted_at)
post_attachments (attachment_id, post_id, url, filename, position, created_at)
post_reactions (user_id, post_id, symbol, created_at)
comments (comment_id, post_id, user_id, parent_comment_id, content, edited_at, created_at, deleted_at)
comment_attachments (attachment_id, comment_id, url, filename, position, created_at)
comment_reactions (user_id, comment_id, symbol, created_at)
badges (badge_id, name, description, category, icon_url, created_at, updated_at)
user_badges (user_id, badge_id, awarded_at)
games (game_id, slug, name, description, thumbnail_url, is_active, base_daily_plays, premium_daily_plays, max_points_per_play, created_at, updated_at)
game_plays (play_id, game_id, user_id, score, points_awarded, played_at)
point_transactions (transaction_id, user_id, type, delta, balance_after, reference_id, note, created_at)
cosmetic_items (cosmetic_id, type, name, description, rarity, render_data, point_cost, created_at)
user_cosmetics (user_id, cosmetic_id, obtained_at, source)
user_equipped_cosmetics (user_id, slot, cosmetic_id)
merch_items (merch_id, name, description, image_url, point_cost, is_available, created_at, updated_at)
merch_variants (variant_id, merch_id, label, stock)

merch_orders (order_id, user_id, points_spent, status, shipping_name, shipping_address, shipping_city, shipping_country, shipping_zip, tracking_number, created_at, updated_at)

merch_order_items (order_item_id, order_id, merch_id, variant_id, quantity, point_cost)

subscriptions (subscription_id, user_id, plan, status, started_at, ends_at, cancelled_at)

subscription_payments (payment_id, subscription_id, amount_cents, currency, payment_provider, provider_payment_id, paid_at)

reports (report_id, reporter_id, target_type, target_id, reason, status, reviewed_by, reviewed_at, resolution_note, created_at)

notifications (notification_id, user_id, type, title, body, entity_type, entity_id, read_at, created_at)

audit_log (log_id, actor_id, action, target_type, target_id, summary, created_at)

appeals (appeal_id, user_id, ban_id, reason, status, reviewed_by, review_note, reviewed_at, created_at)

3. Tabele

3.1. users

Sadrži podatke o korisniku koji se prijavio na platformu.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes		No
username	varchar(32)	No		No
display_name	varchar(80)	No		Yes
email	varchar(254)	No		No
password_hash	text	No		No
is_email_verified	boolean	No		No
google_id	text	No		Yes
github_id	text	No		Yes
github_username	varchar(39)	No		Yes
github_access_token	text	No		Yes
linkedin_id	text	No		Yes
avatar_url	text	No		Yes
banner_url	text	No		Yes
bio	text	No		Yes
last_login_at	timestampz	No		Yes
created_at	timestampz	No		No

updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.2. administrators

Sadrži podatke o administratorima platforme.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
granted_by	uuid	No	administrators	Yes
granted_at	timestampz	No		No

3.3. members

Sadrži podatke o običnom članu platforme.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
points	bigint	No		No

3.4. organizations

Sadrži podatke o organizaciji koja organizuje hakatone na platformi. Atribut verification_status pokazuje status verifikacije organizacije od strane administratora.

org_verification_status = {'pending', 'approved', 'rejected'}

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
name	varchar(100)	No		No
website_url	text	No		Yes
logo_url	text	No		Yes
contact_email	varchar(254)	No		Yes
verification_status	org_verification_status	No		No
reviewed_by	uuid	No	administrators	Yes
reviewed_at	timestampz	No		Yes
rejection_reason	text	No		Yes

3.5. follows

Sadrži podatke o praćenju korisnika (usmerena veza: follower prati followee).

Name	Datatype	PK	FK	Nullable
follower_id	uuid	Yes	users	No
followee_id	uuid	Yes	users	No
created_at	timestampz	No		No

3.6. friendships

Sadrži podatke o prijateljstvima između članova (uzajamna veza). Atribut status označava stanje zahteva za prijateljstvo.

friendship_status = {'pending', 'accepted'}

Name	Datatype	PK	FK	Nullable
user_id_a	uuid	Yes	members	No
user_id_b	uuid	Yes	members	No
requester_id	uuid	No	members	No
status	friendship_status	No		No
requested_at	timestampz	No		No
responded_at	timestampz	No		Yes

3.7. user_bans

Sadrži podatke o korisničkim banovima – sankcijama koje administratori izriču članovima platforme.

Name	Datatype	PK	FK	Nullable
ban_id	uuid	Yes		No
user_id	uuid	No	users	No
banned_by	uuid	No	administrators	No
reason	text	No		No
banned_at	timestampz	No		No
lifted_at	timestampz	No		Yes
lifted_by	uuid	No	administrators	Yes

3.8. user_settings

Sadrži podešavanja privatnosti i obaveštenja za svakog korisnika. Povezana je jedan-na-jedan sa tabelom users.

profile_visibility = {'all', 'members', 'none'}

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
profile_visibility	profile_visibility	No		No
visible_to_recruiters	boolean	No		No
show_email	boolean	No		No
show_location	boolean	No		No
email_notifications	boolean	No		No
push_notifications	boolean	No		No
updated_at	timestampz	No		No

3.9. user_blocks

Sadrži podatke o blokiranju korisnika (usmerena veza: blocker blokira blocked).

Name	Datatype	PK	FK	Nullable
blocker_id	uuid	Yes	users	No
blocked_id	uuid	Yes	users	No
created_at	timestampz	No		No

3.10. skills

Sadrži podatke o veštinama koje članovi mogu posedovati i koje hakatoni mogu zahtevati.

Name	Datatype	PK	FK	Nullable
skill_id	uuid	Yes		No
name	varchar(100)	No		No
category	varchar(50)	No		Yes

3.11. member_skills

Sadrži podatke o veštinama koje poseduju članovi platforme. Predstavlja vezu između tabela members i skills.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	members	No
skill_id	uuid	Yes	skills	No
source	varchar(20)	No		No
verified	boolean	No		No

3.12. hackathons

Sadrži podatke o hackathonima – takmičenjima koja se organizuju na platformi.

hackathon_type = {'physical', 'virtual', 'hybrid'}

hackathon_status = {'upcoming', 'ongoing', 'finished', 'cancelled'}

Name	Datatype	PK	FK	Nullable
hackathon_id	uuid	Yes		No
organization_id	uuid	No	organizations	No
title	varchar(200)	No		No
description	text	No		No
type	hackathon_type	No		No
status	hackathon_status	No		No
theme	varchar(100)	No		Yes
starts_at	timestampz	No		No
ends_at	timestampz	No		No
registration_deadline	timestampz	No		No
voting_opens_at	timestampz	No		Yes
voting_closes_at	timestampz	No		Yes
max_participants	integer	No		Yes
min_team_size	smallint	No		No
max_team_size	smallint	No		No
location	varchar(200)	No		Yes
coordinates	geography(Point, 4326)	No		Yes
logo_url	text	No		Yes
banner_url	text	No		Yes

created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.13. hackathon_drafts

Sadrži nedovršene (radne) verzije hakatona koje organizacija priprema pre objavljivanja. Ceo formular u izradi čuva se u atributu payload kao JSON.

Name	Datatype	PK	FK	Nullable
draft_id	uuid	Yes		No
organization_id	uuid	No	organizations	No
payload	jsonb	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.14. hackathon_required_skills

Sadrži podatke o veštinama koje hakaton zahteva od učesnika. Predstavlja vezu između tabela hackathons i skills.

Name	Datatype	PK	FK	Nullable
hackathon_id	uuid	Yes	hackathons	No
skill_id	uuid	Yes	skills	No

3.15. bounties

Sadrži podatke o sponzorskim izazovima (bounty) u okviru hakatona. Svaki izazov pripada jednom hakatonu i ima sponzora.

Name	Datatype	PK	FK	Nullable
bounty_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
sponsor_name	varchar(100)	No		No
title	varchar(200)	No		No
theme	varchar(100)	No		Yes
description	text	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.16. hackathon_prizes

Sadrži podatke o nagradama koje se dodeljuju u okviru hakatona ili pojedinačnog bounty izazova.

Name	Datatype	PK	FK	Nullable
prize_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
bounty_id	uuid	No	bounties	Yes
sponsor_name	varchar(100)	No		Yes
title	varchar(200)	No		No
description	text	No		Yes
rank	smallint	No		Yes
award_value	text	No		Yes

3.17. teams

Sadrži podatke o timovima koji učestvuju na hakatonu. Svaki tim pripada jednom hakatonu.

Name	Datatype	PK	FK	Nullable
team_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
name	varchar(100)	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.18. team_members

Sadrži podatke o članovima jednog tima i njihovim ulogama (leader ili member).

team_role = {'leader', 'member'}

Name	Datatype	PK	FK	Nullable
team_id	uuid	Yes	teams	No
user_id	uuid	Yes	members	No
role	team_role	No		No
joined_at	timestampz	No		No
left_at	timestampz	No		Yes
deleted_at	timestampz	No		Yes

3.19. team_join_requests

Sadrži zahteve članova za pridruživanje otvorenom timu. Vođa tima prihvata ili odbija zahtev. Atribut status označava stanje zahteva.

status = {'pending', 'accepted', 'declined'}

Name	Datatype	PK	FK	Nullable
request_id	uuid	Yes		No
team_id	uuid	No	teams	No
user_id	uuid	No	members	No
message	text	No		Yes
status	varchar(20)	No		No
created_at	timestamptz	No		No
responded_at	timestamptz	No		Yes
responded_by	uuid	No	users	Yes

3.20. team_invitations

Sadrži pozivnice koje vođa tima šalje pojedinačnim članovima. Pozvani član prihvata ili odbija pozivnicu. Atribut status označava stanje pozivnice.

status = {'pending', 'accepted', 'declined'}

Name	Datatype	PK	FK	Nullable
invitation_id	uuid	Yes		No
team_id	uuid	No	teams	No
user_id	uuid	No	members	No
invited_by	uuid	No	users	Yes
message	text	No		Yes
status	varchar(20)	No		No
created_at	timestamptz	No		No
responded_at	timestamptz	No		Yes

3.21. applications

Sadrži podatke o prijavama članova na hakatone. Atribut status označava trenutni status prijave.

application_status = {'pending', 'approved', 'rejected', 'waitlisted', 'withdrawn'}

Name	Datatype	PK	FK	Nullable
application_id	uuid	Yes		No
user_id	uuid	No	members	No
hackathon_id	uuid	No	hackathons	No
team_id	uuid	No	teams	Yes
status	application_status	No		No
reviewed_by	uuid	No	users	Yes
reviewed_at	timestampz	No		Yes
rejection_reason	text	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.22. application_questions

Sadrži prilagođena pitanja koja organizator dodaje na prijavni formular hakatona. Atribut type određuje tip polja za unos.

type = {'short_text', 'long_text', 'single_choice', 'multi_choice'}

Name	Datatype	PK	FK	Nullable
question_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
prompt	text	No		No
type	varchar(20)	No		No
options	jsonb	No		Yes
required	boolean	No		No
allow_other	boolean	No		No
position	integer	No		No
created_at	timestampz	No		No

3.23. question_answers

Sadrži odgovore prijavljenog člana na pojedinačna pitanja prijavnog formulara hakatona.

Name	Datatype	PK	FK	Nullable
answer_id	uuid	Yes		No
application_id	uuid	No	applications	No
question_id	uuid	No	application_questions	No
answer	text	No		No
created_at	timestampz	No		No

3.24. projects

Sadrži podatke o projektima koje timovi razvijaju u okviru hakatona. Atribut status opisuje trenutnu fazu projekta.

project_status = {'draft', 'submitted', 'under_review', 'judged'}

Name	Datatype	PK	FK	Nullable
project_id	uuid	Yes		No
team_id	uuid	No	teams	No
status	project_status	No		No
title	varchar(200)	No		No
description	text	No		Yes
repository_url	text	No		Yes
video_url	text	No		Yes
submitted_at	timestampz	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.25. hackathon_results

Sadrži podatke o rezultatima projekata na hakatonu i pripadajućim nagradama. Ako je bounty_id NULL, rezultat se odnosi na generalno rangiranje.

Name	Datatype	PK	FK	Nullable
result_id	uuid	Yes		No
project_id	uuid	No	projects	No
bounty_id	uuid	No	bounties	Yes
rank	integer	No		Yes
prize_id	uuid	No	hackathon_prizes	Yes

3.26. bounty_submissions

Sadrži veze između projekata i bounty izazova na koje su projekti predati.

Name	Datatype	PK	FK	Nullable
bounty_id	uuid	Yes	bounties	No
project_id	uuid	Yes	projects	No

3.27. votes

Sadrži glasove publike za projekte (nagrada "People's Choice"). Glasač može biti prijavljeni korisnik (voter_id) ili gost identifikovan preko voter_fingerprint.

Name	Datatype	PK	FK	Nullable
vote_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
project_id	uuid	No	projects	No
voter_id	uuid	No	users	Yes
voter_fingerprint	text	No		Yes
created_at	timestampz	No		No

3.28. kanban_boards

Sadrži podatke o Kanban tabli koju koristi tim za vođenje projekta. Svaki tim ima najviše jednu tablu.

Name	Datatype	PK	FK	Nullable
board_id	uuid	Yes		No
team_id	uuid	No	teams	No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.29. kanban_columns

Sadrži podatke o kolonama na Kanban tabli.

Name	Datatype	PK	FK	Nullable
column_id	uuid	Yes		No

board_id	uuid	No	kanban_boards	No
name	varchar(100)	No		No
position	real	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.30. kanban_cards

Sadrži podatke o pojedinačnim karticama (zadacima) na Kanban tabli.

Name	Datatype	PK	FK	Nullable
card_id	uuid	Yes		No
column_id	uuid	No	kanban_columns	No
created_by	uuid	No	members	No
assigned_to	uuid	No	members	Yes
title	varchar(200)	No		No
description	text	No		Yes
position	real	No		No
due_at	timestampz	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.31. servers

Sadrži podatke o serveru za komunikaciju koji se vezuje za hakaton (slično Discord serveru).

Name	Datatype	PK	FK	Nullable
server_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
name	varchar(200)	No		No
logo_url	text	No		Yes
banner_url	text	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.32. server_roles

Sadrži podatke o ulogama koje postoje na nekom serveru za komunikaciju.

Name	Datatype	PK	FK	Nullable
server_role_id	uuid	Yes		No
server_id	uuid	No	servers	No
name	varchar(100)	No		No
created_at	timestampz	No		No

3.33. permissions

Sadrži podatke o dozvolama koje se mogu dodeliti ulogama na serveru.

Name	Datatype	PK	FK	Nullable
permission_id	uuid	Yes		No
name	varchar(100)	No		No
description	text	No		No

3.34. server_role_permissions

Sadrži vezu između uloga i dozvola – koje dozvole su dodeljene kojoj ulozi.

Name	Datatype	PK	FK	Nullable
server_role_id	uuid	Yes	server_roles	No
permission_id	uuid	Yes	permissions	No

3.35. user_roles

Sadrži podatke o tome koja uloga je dodeljena kom korisniku na serveru.

Name	Datatype	PK	FK	Nullable
server_role_id	uuid	Yes	server_roles	No
user_id	uuid	Yes	members	No
assigned_by	uuid	No	users	Yes
assigned_at	timestampz	No		No

3.36. channel_groups

Sadrži podatke o grupama kanala (kategorijama) u okviru servera. Atribut position definiše redosled.

Name	Datatype	PK	FK	Nullable
group_id	uuid	Yes		No
server_id	uuid	No	servers	No
name	varchar(100)	No		No
position	real	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.37. channels

Sadrži podatke o kanalima na serveru. Kanal može biti opšti, za obaveštenja, timski ili privatni.

channel_type = {'general', 'announcements', 'team', 'private', 'project', 'kanban'}

Name	Datatype	PK	FK	Nullable
channel_id	uuid	Yes		No
group_id	uuid	No	channel_groups	No
team_id	uuid	No	teams	Yes
type	channel_type	No		No
name	varchar(100)	No		No
position	real	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.38. channel_members

Sadrži listu korisnika sa pristupom privatnom kanalu (kontrola pristupa privatnih kanala).

Name	Datatype	PK	FK	Nullable
channel_id	uuid	Yes	channels	No
user_id	uuid	Yes	users	No
added_by	uuid	No	users	Yes
added_at	timestampz	No		No

3.39. channel_pins

Sadrži zakačene (pinovane) poruke u okviru kanala.

Name	Datatype	PK	FK	Nullable
channel_id	uuid	Yes	channels	No
message_id	uuid	Yes	messages	No
pinned_by	uuid	No	users	Yes
pinned_at	timestampz	No		No

3.40. messages

Sadrži podatke o porukama koje korisnici razmenjuju (kako u kanalima, tako i u direktnim razgovorima). Atribut reply_to_id omogućava odgovor na poruku.

Name	Datatype	PK	FK	Nullable
message_id	uuid	Yes		No
sender_id	uuid	No	users	No
reply_to_id	uuid	No	messages	Yes
content	text	No		No
sent_at	timestampz	No		No
edited_at	timestampz	No		Yes
deleted_at	timestampz	No		Yes
deleted_by	uuid	No	users	Yes

3.41. message_attachments

Sadrži podatke o priložima (slike, dokumenti) koji su priloženi uz poruku.

Name	Datatype	PK	FK	Nullable
attachment_id	uuid	Yes		No
message_id	uuid	No	messages	No
url	text	No		No
filename	varchar(255)	No		Yes
position	smallint	No		No
created_at	timestampz	No		No

3.42. channel_messages

Veza koja označava da je poruka poslata u određeni kanal servera.

Name	Datatype	PK	FK	Nullable
message_id	uuid	Yes	messages	No
channel_id	uuid	No	channels	No

3.43. conversations

Sadrži podatke o privatnim konverzacijama (direktnim porukama) između korisnika.

Name	Datatype	PK	FK	Nullable
conversation_id	uuid	Yes		No
created_by	uuid	No	users	No
name	varchar(100)	No		Yes
icon	varchar(512)	No		Yes
created_at	timestampz	No		No

3.44. conversation_members

Sadrži podatke o učesnicima konverzacije (jedan-na-jedan ili grupne direktne poruke).

Name	Datatype	PK	FK	Nullable
conversation_id	uuid	Yes	conversations	No
user_id	uuid	Yes	users	No
joined_at	timestampz	No		No
left_at	timestampz	No		Yes
last_read_at	timestampz	No		Yes

3.45. direct_messages

Veza koja označava da je poruka deo privatne konverzacije, a ne kanala servera.

Name	Datatype	PK	FK	Nullable
message_id	uuid	Yes	messages	No
conversation_id	uuid	No	conversations	No

3.46. message_reactions

Sadrži reakcije korisnika (emodži) na poruke. Atribut symbol predstavlja emodži.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
message_id	uuid	Yes	messages	No
symbol	text	Yes		No
created_at	timestampz	No		No

3.47. server_mutes

Sadrži podatke o utišavanju (mute) korisnika na serveru hakatona, sa opcionim rokom isteka (expires_at).

Name	Datatype	PK	FK	Nullable
mute_id	uuid	Yes		No
server_id	uuid	No	servers	No
muted_user_id	uuid	No	users	No
muted_by	uuid	No	users	Yes
muted_at	timestampz	No		No
expires_at	timestampz	No		Yes
reason	text	No		Yes

3.48. posts

Sadrži podatke o objavama koje korisnici postavljaju na društvenom feedu platforme.

Name	Datatype	PK	FK	Nullable
post_id	uuid	Yes		No
user_id	uuid	No	users	No
content	text	No		No
created_at	timestampz	No		No
edited_at	timestampz	No		Yes
deleted_at	timestampz	No		Yes

3.49. post_attachments

Sadrži podatke o priložima (slike, dokumenti) priloženim uz objavu.

Name	Datatype	PK	FK	Nullable
attachment_id	uuid	Yes		No
post_id	uuid	No	posts	No

url	text	No		No
filename	varchar(255)	No		Yes
position	smallint	No		No
created_at	timestampz	No		No

3.50. post_reactions

Sadrži reakcije korisnika (emodži) na objave.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
post_id	uuid	Yes	posts	No
symbol	text	Yes		No
created_at	timestampz	No		No

3.51. comments

Sadrži podatke o komentarima ostavljenim na objavama. Atribut parent_comment_id omogućava ugnježdene odgovore.

Name	Datatype	PK	FK	Nullable
comment_id	uuid	Yes		No
post_id	uuid	No	posts	No
user_id	uuid	No	users	No
parent_comment_id	uuid	No	comments	Yes
content	text	No		No
edited_at	timestampz	No		Yes
created_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.52. comment_attachments

Sadrži podatke o priložima priloženim uz komentar.

Name	Datatype	PK	FK	Nullable
attachment_id	uuid	Yes		No
comment_id	uuid	No	comments	No
url	text	No		No

filename	varchar(255)	No		Yes
position	smallint	No		No
created_at	timestampz	No		No

3.53. comment_reactions

Sadrži reakcije korisnika (emodži) na komentare.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
comment_id	uuid	Yes	comments	No
symbol	text	Yes		No
created_at	timestampz	No		No

3.54. badges

Sadrži podatke o bedževima koji se dodeljuju članovima.

badge_category = {'participation', 'achievement', 'social', 'special'}

Name	Datatype	PK	FK	Nullable
badge_id	uuid	Yes		No
name	varchar(100)	No		No
description	text	No		No
category	badge_category	No		No
icon_url	text	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.55. user_badges

Sadrži podatke o bedževima koje je član osvojio i datumu dodele.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	members	No
badge_id	uuid	Yes	badges	No
awarded_at	timestampz	No		No

3.56. games

Sadrži podatke o dnevnim mini-igrama dostupnim na platformi.

Name	Datatype	PK	FK	Nullable
game_id	uuid	Yes		No
slug	varchar(50)	No		No
name	varchar(100)	No		No
description	text	No		Yes
thumbnail_url	text	No		Yes
is_active	boolean	No		No
base_daily_plays	smallint	No		No
premium_daily_plays	smallint	No		No
max_points_per_play	integer	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.57. game_plays

Sadrži rezultate odigranih mini-igara i služi kao osnova za rang-listu (leaderboard).

Name	Datatype	PK	FK	Nullable
play_id	uuid	Yes		No
game_id	uuid	No	games	No
user_id	uuid	No	members	No
score	integer	No		No
points_awarded	integer	No		No
played_at	timestampz	No		No

3.58. point_transactions

Sadrži knjigu (ledger) svih promena stanja poena članova. Zapisi se samo dodaju (append-only). Atribut type označava razlog transakcije.

```
point_txn_type = {'game_reward', 'badge_award',
                  'hackathon_placement', 'bounty_placement', 'merch_purchase',
                  'premium_purchase', 'admin_adjustment'}
```

Name	Datatype	PK	FK	Nullable
transaction_id	uuid	Yes		No
user_id	uuid	No	members	No
type	point_txn_type	No		No
delta	bigint	No		No
balance_after	bigint	No		No
reference_id	uuid	No		Yes
note	text	No		Yes
created_at	timestampz	No		No

3.59. cosmetic_items

Sadrži podatke o kozmetičkim predmetima (efekti za korisničko ime, dekoracije avatara, efekti banera).

cosmetic_type = {'username_effect', 'avatar_decoration', 'banner_effect'}

cosmetic_rarity = {'common', 'rare', 'epic', 'legendary'}

Name	Datatype	PK	FK	Nullable
cosmetic_id	uuid	Yes		No
type	cosmetic_type	No		No
name	varchar(100)	No		No
description	text	No		Yes
rarity	cosmetic_rarity	No		No
render_data	jsonb	No		No
point_cost	integer	No		Yes
created_at	timestampz	No		No

3.60. user_cosmetics

Sadrži podatke o kozmetičkim predmetima koje je član stekao. Atribut source označava način sticanja.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	members	No
cosmetic_id	uuid	Yes	cosmetic_items	No
obtained_at	timestampz	No		No
source	varchar(50)	No		No

3.61. user_equipped_cosmetics

Sadrži podatke o trenutno postavljenim kozmetičkim predmetima člana po slotu (tipu).

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	members	No
slot	cosmetic_type	Yes		No
cosmetic_id	uuid	No	cosmetic_items	No

3.62. merch_items

Sadrži podatke o merchandise predmetima koji se mogu kupiti za poene. Atribut is_available pokazuje da li je predmet trenutno dostupan.

Name	Datatype	PK	FK	Nullable
merch_id	uuid	Yes		No
name	varchar(200)	No		No
description	text	No		Yes
image_url	text	No		Yes
point_cost	integer	No		No
is_available	boolean	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.63. merch_variants

Sadrži podatke o varijantama merchandise predmeta (npr. veličina, boja) i njihovim stanjima na zalihama.

Name	Datatype	PK	FK	Nullable
variant_id	uuid	Yes		No
merch_id	uuid	No	merch_items	No
label	varchar(100)	No		No
stock	integer	No		No

3.64. merch_orders

Sadrži podatke o porudžbinama merchandise predmeta i adresi za isporuku. Atribut `points_spent` predstavlja broj utrošenih poena.

`merch_order_status` = {'pending', 'processing', 'shipped', 'delivered', 'cancelled'}

Name	Datatype	PK	FK	Nullable
order_id	uuid	Yes		No
user_id	uuid	No	members	No
points_spent	integer	No		No
status	merch_order_status	No		No
shipping_name	varchar(200)	No		No
shipping_address	text	No		No
shipping_city	varchar(100)	No		No
shipping_country	char(2)	No		No
shipping_zip	varchar(20)	No		No
tracking_number	varchar(100)	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.65. merch_order_items

Sadrži podatke o pojedinačnim stavkama u okviru jedne merchandise porudžbine. Atribut `point_cost` je cena u poenima u trenutku porudžbine.

Name	Datatype	PK	FK	Nullable
order_item_id	uuid	Yes		No
order_id	uuid	No	merch_orders	No
merch_id	uuid	No	merch_items	No
variant_id	uuid	No	merch_variants	Yes
quantity	smallint	No		No
point_cost	integer	No		No

3.66. subscriptions

Sadrži podatke o premium pretplatama korisnika.

`subscription_plan` = {'premium'}

subscription_status = {'active', 'cancelled', 'expired'}

Name	Datatype	PK	FK	Nullable
subscription_id	uuid	Yes		No
user_id	uuid	No	members	No
plan	subscription_plan	No		No
status	subscription_status	No		No
started_at	timestampz	No		No
ends_at	timestampz	No		No
cancelled_at	timestampz	No		Yes

3.67. subscription_payments

Sadrži podatke o pojedinačnim uplatama vezanim za pretplate. Atribut amount_cents je iznos u centima izabrane valute.

Name	Datatype	PK	FK	Nullable
payment_id	uuid	Yes		No
subscription_id	uuid	No	subscriptions	No
amount_cents	integer	No		No
currency	char(3)	No		No
payment_provider	varchar(50)	No		No
provider_payment_id	text	No		No
paid_at	timestampz	No		No

3.68. reports

Sadrži podatke o prijavama neprikladnog sadržaja ili neregularnosti. Atribut target_type i target_id čine polimorfnu referencu na prijavljenu jedinicu.

report_target_type = {'user', 'post', 'comment', 'message', 'hackathon'}

report_status = {'pending', 'reviewed', 'resolved', 'dismissed'}

Name	Datatype	PK	FK	Nullable
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report_id	uuid	Yes		No
reporter_id	uuid	No	users	No
target_type	report_target_type	No		No
target_id	uuid	No		No
reason	text	No		No
status	report_status	No		No
reviewed_by	uuid	No	administrators	Yes
reviewed_at	timestampz	No		Yes
resolution_note	text	No		Yes
created_at	timestampz	No		No

3.69. notifications

Sadrži podatke o obaveštenjima koja se prikazuju korisnicima. Atributi entity_type i entity_id čine polimorfnu referencu na povezan entitet.

```
notification_type = {'application_approved', 'application_rejected',
'application_waitlisted', 'badge_awarded', 'hackathon_result_posted',
'hackathon_starting_soon', 'organization_verified',
'organization_rejected', 'new_direct_message', 'position_assigned',
'bounty_result_posted', 'merch_order_shipped', 'new_follower',
'friend_request_received', 'friend_request_accepted',
'team_invitation_received', 'team_request_received',
'team_request_accepted', 'post_comment', 'post_reaction', 'mention',
'new_application'}
```

```
entity_type = {'user', 'hackathon', 'application', 'team', 'project', 'post',
'comment', 'badge', 'message', 'bounty', 'game'}
```

Name	Datatype	PK	FK	Nullable
notification_id	uuid	Yes		No
user_id	uuid	No	users	No
type	notification_type	No		No
title	varchar(100)	No		No
body	text	No		Yes
entity_type	entity_type	No		Yes
entity_id	uuid	No		Yes
read_at	timestampz	No		Yes
created_at	timestampz	No		No

3.70. audit_log

Sadrži zapise o moderatorskim i administratorskim akcijama na platformi.

Name	Datatype	PK	FK	Nullable
log_id	uuid	Yes		No
actor_id	uuid	No	administrators	Yes
action	varchar(64)	No		No
target_type	varchar(32)	No		Yes
target_id	uuid	No		Yes
summary	text	No		No
created_at	timestampz	No		No

3.71. appeals

Sadrži žalbe korisnika na izrečene banove. Atribut status označava stanje žalbe.

appeal_status = {'pending', 'approved', 'rejected'}

Name	Datatype	PK	FK	Nullable
appeal_id	uuid	Yes		No
user_id	uuid	No	users	No
ban_id	uuid	No	user_bans	Yes
reason	text	No		No
status	appeal_status	No		No
reviewed_by	uuid	No	administrators	Yes
review_note	text	No		Yes
reviewed_at	timestampz	No		Yes
created_at	timestampz	No		No

Verzije dokumenta

Verzija	Datum	Opis	Autori
Alfa	17.05.2026.	Alfa verzija	Andrej Čolić
1.0	21.05.2026.	Kompletna verzija	Andrej Čolić
1.1	14.07.2026.	Unapređena verzija	Andrej Čolić